FORMOSA CRYPTO

# Completing the Chain:

## Verified Implementations of Hash-Based Signatures and Their Security

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# Jasmin [ABB+17, ABB+20]

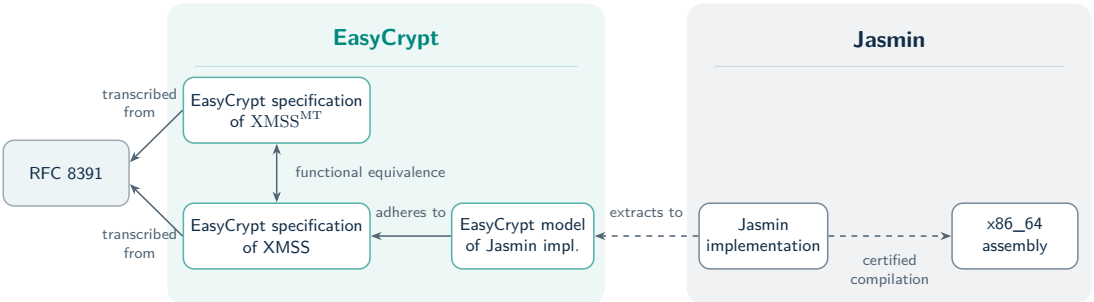
# Jasmin [ABB+17, ABB+20]

- Stack zeroization [OBG+24]
- (Static) Safety checker based on abstract interpretation
- Compiler correctness proved in Rocq
- Extraction to EasyCrypt
- Speculative constant-time:
  - Spectre v1 [SBG+23] and other spectre variants [ABC+25]
  - Compilation preserves SCT [ABB+25].

**Compiler Correctness & Constant-Time Preservation**

Compilation preserves the semantics of *safe* source programs and constant-time behavior: functional correctness and constant-time guarantees established at source level carry to the assembly level.

# Verification Workflow



# Trusted Computing Base

- Correctness of the SHA-256 implementation is outside the scope of the proof

```
op Hash : W8.t list -> nbytes. (* Abstract operator *)

axiom hash_128 (x : W8.t Array128.t):
  phoare[
    M(Syscall).hash_plen_3n____sha256:
    arg.`2 = x
    ==>
    to_list res = NBytes.val (Hash (to_list x))
  ] = 1%r.
```

Figure: EasyCrypt axiom stating the correctness of SHA-256.

# Trusted Computing Base

- Jasmin compiler: unverified compilation passes (e.g. parsing, pretyping, etc.)
- Jasmin extraction to EasyCrypt
- SMT Solvers (Z3, CVC4 and Alt-Ergo)

# Implementation: Performance

| | C Ref | Jasmin (abs, %) | loop zeroization (abs, %) | unrolled zeroization (abs, %) |
|---|---|---|---|---|
| **Keygen** | 10092.74 | 8263.54 (-18.12%) | 8293.61 (+0.36%) | 8312.35 (+0.59%) |
| **Sign** | 20251.01 | 24775.17 (+22.34%) | 24872.05 (+0.39%) | 24927.53 (+0.61%) |
| **Verify** | 10.35 | 8.47 (-18.14%) | 8.48 (+0.13%) | 8.95 (+5.69%) |

Table: Performance Comparison between C and Jasmin implementations in $10^6$ clock cycles (column #2). Zeroization overhead w.r.t. Jasmin baseline (columns #3 and #4).

# QUESTIONS?

# References I


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